Improving Positioning Accuracy Using GPS Pseudorange Measurements for Cooperative Vehicular Localization

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***Abstract*—Accurate positioning is a key factor for enabling innovative applications in intelligent transportation systems. Cutting-edge communication technologies make cooperative localization a promising approach for accurate vehicle positioning. In this paper, we first propose a ranging technique called *weighted least squares double difference* (WLS-DD), which is used to detect intervehicle distances based on the sharing of GPS pseudorange measurements and a weighted least squares method. It takes the carrier-to-noise ratio (CNR) of raw pseudorange measurements into consideration for mitigating noises so that it can improve the accuracy of the distance detection. We show the superiority of WLS-DD by conducting a series of field experiments. Based on intervehicle distance detection, we propose a *distributed location estimate algorithm* (DLEA) to improve the accuracy of vehicle positioning. The implementation of DLEA only relies on inaccurate GPS pseudorange measurements and the obtained intervehicle distances without using any reference points for positioning correction. Moreover, to evaluate the joint effect of WLS-DD and DLEA, we derive a data fitting model based on the observed distance detection bias from field experiments, which generates parameters in a variety of environments for performance evaluation. Finally, we demonstrate the effectiveness of the proposed solutions via a comprehensive simulation study.**

***Index Terms*—Cooperative localization, global positioning system (GPS), ranging technique, vehicular networks.**

I. Introduction

RECENT advances in wireless communications encourage the development of cooperative positioning systems in vehicular networks. Accurate positioning is one of the most vital premises for the implemention of emerging intelligent transportation systems, such as collision warning systems [1], speed advisory systems [2], and road reservation systems [3], etc. Unfortunately, current commercially available GPS devices suffer from inaccuracy to varying degrees. Indeed, a standalone GPS device typically suffers the positioning error up to tens of meters [4], which is too far for many safety-critical applications.

Extensive studies have been devoted to improving vehicle positioning accuracy. Incorporating GPS with auxiliary information (e.g., ranging measurements, reference points, and digital maps, etc.) is a primary approach to enhancing the system performance. A number of solutions are proposed for improving positioning accuracy based on the measurement of intervehicle distances [5]–[7]. Nevertheless, in one way or another, these technologies rely on radio-based ranging methods, such as time of arrival (ToA), time difference of arrival (TDoA), angle of arrival (AoA), and received signal strength, which normally require particular sensors with high computational complexity or suffer from inadequate ranging accuracy [8]. Leveraging reference points (e.g., anchors or base stations) with known exact positions is another prevalent approach [9]–[11]. The underlying principle of these technologies is inspired from differential GPS (DGPS) [12]. In general, these systems assume that the GPS receivers are relatively close to the reference point and have similar atmospheric errors. Accordingly, the reference point can be used to compute the GPS bias for positioning correction. Although the given solutions have been demonstrated as effective in improving the positioning accuracy, they either rely on complex techniques with high computational cost (e.g., ToA and TDoA) or require particular hardware devices (e.g., sensors and reference points), which make them hard to deploy. In this paper, we propose new solutions that complement previous work with low implementation cost. In particular, the proposed solutions are expected to be most effective in open environments such as highways, suburban areas, and rural areas, etc. Nevertheless, in other environments, solutions for compensating GPS outages and overcoming GPS limitations can be found in [5], [6], and [13].

It has been demonstrated that dedicated short-range communication (DSRC) is a superior standard in supporting high-speed and low-latency data dissemination in vehicular networks. The DSRC radio technology is being standardized as IEEE 802.11p, and the overall DSRC communication stack

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is being standardized by the IEEE 1609 working group [14]. DSRC supports data rates from 3 to 27 Mb/s with multichannel operation, and it supports the communication range up to 300 m in practice. Clearly, DSRC is the most appealing solution for enabling cooperative vehicular localization systems.

The main contributions of this paper are summarized as follows. First, we propose a ranging technique called *weighted least squares double difference* (WLS-DD), which is based on the sharing of GPS pseudorange measurements and a weighted least squares method. The carrier-to-noise ratio (CNR) of raw pseudorange measurements are taken into account for noise mitigation. The field experiments show that WLS-DD can improve the accuracy of distance detection significantly. Second, we propose a *distributed location estimate algorithm* (DLEA), which estimates vehicle locations based on the sharing of pesudorange measurements and the detection of intervehicle distances among neighboring vehicles. Moreover, we show the feasibility of adopting DSRC to enable cooperative positioning by analyzing the communication overhead. Third, to examine the effect of intervehicle distance detection on the performance of vehicle location estimate, we derive a data fitting model, which fits the observed errors of distance detection from filed experiments into an appropriate probability distribution. Finally, based on the parameters derived from the data fitting model, we conduct an extensive simulation study, and the results demonstrate the superiority of the proposed solutions. Note that the current solutions are only evaluated in static scenarios. Nevertheless, more advanced techniques such as map mapping, dead reckoning, sensor fusion, etc., can be incorporated to further enhance the robustness of the positioning system.

The remainder of this paper is organized as follows. Section II reviews the related work. In Section III, we propose a ranging technique and conduct a series of field experiments to demonstrate the effectiveness of the proposed solution. In Section IV, we propose a cooperative localization algorithm and justify its feasibility. Section V illustrates a data fitting model and demonstrates the suitability of the derived parameters for performance evaluation and Section VI gives a comprehensive simulation study. Finally, Section VII concludes this paper and discusses future research directions.

II. Related Work

A significant amount of research has been devoted to vehicular positioning. The fusion of GPS measurements with auxiliary information, such as onboard sensors, vehicle kinematics, and digital road maps, etc., is one of the primary approaches [4]. Lee *et al.* [9] proposed a mobile version of the DGPS system, which exploits the RFID technique in vehicle positioning. The basic idea is that, when a vehicle obtains a precise location from an RFID tag installed along the road, it acts as a moving reference point temporarily to correct the position of its neighboring vehicles. Jo *et al.* [15] developed a positioning system by integrating GPS measurements, in-vehicle sensors, and onboard navigation systems so that the positioning can adapt to various driving conditions. With the development of geographic information systems, the availability of accurate geographic data is used to further mitigate GPS errors. *Map matching* is a representative technique for improving positioning accuracy [16]. Suganuma and Uozumi [17] proposed an algorithm based on map matching to compensate the drift error of measurements from the hybrid of a global navigation satellite system and an inertial navigation system (GNSS/INS). The algorithm estimates the GNSS/INS drift error using digital maps and position measurements from onboard sensors.

A number of techniques such as dead reckoning, cellular localization, image/video localization, etc., have been proposed to overcome GPS limitations [5], [6], [13]. Vu *et al.* [18] proposed a sensor fusion technique to aid the INS by using computer vision and DGPS measurements. Image features of the traffic light are captured by a forward-looking camera and applied to assist the positioning. Kloeden *et al.* [19] proposed a localization system based on cooperative RF-based landmarks. An onboard sensor allows the vehicle to measure the round-trip time of flight and the AoA of the incoming RF wave, which is applied to determine the distance and the angle between the landmark and the vehicle. With the known landmark coordinates, an extended-Kalman-filter-based approach is proposed to estimate the vehicle position. Parker and Valaee [7] proposed a cooperative positioning algorithm based on intervehicle distance detection. The signal-strength-based radio ranging technique is applied to measure the distance between vehicles. In addition, both road map information and vehicle kinematics are incorporated to estimate relative positions of vehicles. These existing technologies can improve positioning accuracy to varying degrees, but they heavily rely on the support of particular hardware devices or large-scale infrastructures, whereas some of them are not yet globally available. Unlike previous studies, the proposed solutions in this paper neither rely on conventional radio-based ranging techniques nor require any particular ranging sensors. Instead, the inaccurate GPS pseudorange measurements and the corresponding CNRs are exploited for cooperative vehicular localization.

III. New Ranging Technique

We propose a ranging technique to detect intervehicle distances.

A. Classic Double Difference Pseudorange Solution

In the GPS system, the distance between a vehicle and a satellite is derived from the estimated time of signal transmission, and this distance is called the *pseudorange*. Several sources contribute to the errors in pseudorange measurements. Specifically, the pseudorange between GPS receiver a and satellite i, which is denoted PR_a^i, can be decomposed into [12]

$$\mathrm{PR}_a^i = R_a^i + t_a + x^i + \varepsilon_a^i \tag{1}$$

where R_a^i is the true distance between satellite i and receiver a; t_a is the error caused by receiver a's clock bias; x^i is the common noise related to satellite i, including satellite clock bias, atmospheric delay, and errors in the broadcast ephemeris;

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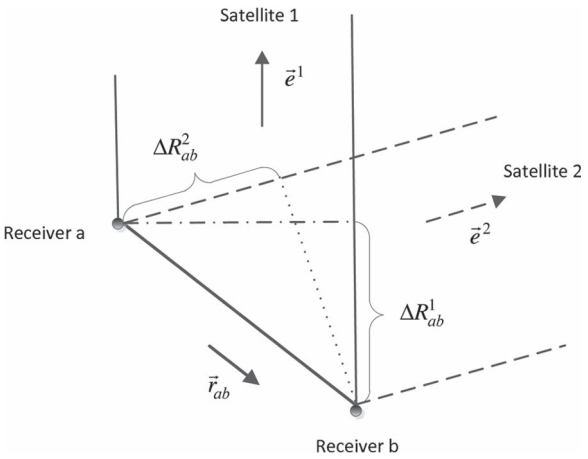


Fig. 1. Pseudorange double difference.

and ε^i_a is the noncommon noise related to both receiver a and satellite i, including multipath and code acquisition noise. By taking the difference between the pseudoranges of two receivers a and b to the same satellite i, the common noise due to satellite i can be removed as follows:

$$\begin{aligned} S^i_{ab} &= \mathrm{PR}^i_a - \mathrm{PR}^i_b \\ &= \Delta R^i_{ab} + (t_a - t_b) + \left(\varepsilon^i_a - \varepsilon^i_b\right) \end{aligned} \tag{2}$$

where S^i_{ab} is the *single difference* of pseudorange measurements and $\Delta R^i_{ab} = (R^i_a - R^i_b)$, which is the difference between the true ranges from receivers a and b to satellite i. As the true ranges from satellite i to receivers a and b are much larger than the distance between a and b, the two vectors pointing from i to a and i to b are nearly parallel to each other. As shown in Fig. 1, ΔR^i_{ab} can be thus approximated by

$$\Delta R^i_{ab} = \vec{e}^i \cdot \vec{r}_{ab} \tag{3}$$

where $\vec{e}^i$ is the unit vector pointing from receiver a (or b) to satellite i, and $\vec{r}_{ab}$ is the distance vector between receivers a and b. When two satellites i and j are available to both of the receivers, the clock bias of a and b can be further removed by taking the *double difference* based on (2) and (3) as follows:

$$\begin{aligned} D^{ij}_{ab} &= S^i_{ab} - S^j_{ab} \\ &= [\vec{e}^i - \vec{e}^j] \cdot \vec{r}_{ab} + \left[\left(\varepsilon^i_a - \varepsilon^i_b\right) - \left(\varepsilon^j_a - \varepsilon^j_b\right)\right]. \end{aligned} \tag{4}$$

Given a set of satellites $\{0, 1, \ldots, n\}$ shared by the receivers a and b, and suppose satellite 0 is selected as the *reference satellite* for double difference [i.e., $j = 0$ in (4)], then (4) can be generalized as

$$\mathbf{D}_{ab} = \mathbf{H}\vec{r}_{ab} + \boldsymbol{\varepsilon} \tag{5}$$

where $\mathbf{D}_{ab}$ is the column vector of pseudorange double differences with respect to the satellites 0 and i $(1 \le i \le n)$ ($\mathbf{D}_{ab} = [D^{10}_{ab}\ D^{20}_{ab}\ \cdots\ D^{n0}_{ab}]^T$), $\mathbf{H}$ is the column vector of the difference between two unit vectors ($\mathbf{H} = [(\vec{e}^1 - \vec{e}^0)\ (\vec{e}^2 - \vec{e}^0)\ \cdots\ (\vec{e}^n - \vec{e}^0)]^T$), and $\boldsymbol{\varepsilon}$ is the column vector of aggregated noncommon noise ($\boldsymbol{\varepsilon} = [((\varepsilon^1_a - \varepsilon^1_b) - (\varepsilon^0_a - \varepsilon^0_b)) \ldots ((\varepsilon^n_a - \varepsilon^n_b) - (\varepsilon^0_a - \varepsilon^0_b))]^T$). Assume that $\boldsymbol{\varepsilon}$ is zero mean and equal variance, then $\vec{r}_{ab}$ can be approximately solved by the linear least squares estimator, i.e.,

$$\vec{r}_{ab} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{D}_{ab}. \tag{6}$$

Comparing the distance around 20 000 km from the satellites to the GPS receivers, the positioning error by the GPS, which is on the order of tens of meters, can be ignored when computing the unit vector of $\vec{e}^i$ $(0 \le i \le n)$. Thus, $\vec{e}^i$ can be obtained by the GPS fixes and the corresponding ephemeris. Note that it requires at least four shared satellites to both of the GPS receivers for distance detection.

B. Weighted Least Squares Method

In (5), $\boldsymbol{\varepsilon}$ is the aggregated noncommon noise of pseudorange measurements, including both multipath and random code acquisition errors. The ith item of $\boldsymbol{\varepsilon}$ represents the noncommon noise when computing the double difference using satellite i $(1 \le i \le n)$ and satellite 0 (i.e., the reference satellite), which is

$$\varepsilon^i = \left(\varepsilon^i_a - \varepsilon^i_b\right) - \left(\varepsilon^0_a - \varepsilon^0_b\right). \tag{7}$$

In scenarios where the multipath effect is not severe, such as highways, suburbs, rural areas, etc., $\boldsymbol{\varepsilon}$ is dominated by code acquisition errors, which can be directly linked to the CNR of the received satellite signal. To mitigate the impact of $\boldsymbol{\varepsilon}$, it is straightforward to select the pseudorange measurements with high CNR values for double difference processing. In our previous work [20], we tested this method by setting different thresholds for selecting satellites with a good signal. It has been demonstrated that, although choosing satellites by setting a high CNR threshold could improve the accuracy, the number of valid samples of pseudorange measurements would be reduced significantly. Lack of valid samples for double difference processing will result in substantial delays between successive distance detections, which is not tolerable in highly dynamic vehicular networks.

Therefore, it is expected to improve the accuracy without causing unacceptable delays in successive distance detections. In view of this, we propose the WLS-DD ranging technique. With a nonequal variance of each ε^i, the weighted least squares estimator, i.e.,

$$\vec{r}_{ab} = (\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{W}\mathbf{D}_{ab} \tag{8}$$

can be used as the best linear unbiased estimator, where weight matrix $\mathbf{W}$ is the inverse of the covariance matrix of $\boldsymbol{\varepsilon}$. The detailed operations of WLS-DD are presented as follows.

1) Selection of the Reference Satellite: As shown in (7), given the reference satellite 0, if $(\varepsilon^0_a - \varepsilon^0_b)$ can be kept small, then ε^i can be approximated as follows:

$$\varepsilon^i \approx \left(\varepsilon^i_a - \varepsilon^i_b\right). \tag{9}$$

With independent and zero-mean random variables of ε^0_a and ε^0_b, (9) makes sense only if the standard deviations of both ε^0_a and ε^0_b are kept small. Therefore, the reference satellite

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is expected to have high CNR values to both GPS receivers. Specifically, denoting ϕ_a^0 and ϕ_b^0 as the CNR values from satellite 0 to receivers a and b, respectively, it requires that $\phi_a^0 \geq \mathrm{CNR}_{\mathrm{ref}}$ and $\phi_b^0 \geq \mathrm{CNR}_{\mathrm{ref}}$, where $\mathrm{CNR}_{\mathrm{ref}}$ is the threshold. In case that there are multiple satellites satisfying this condition, one of them are randomly selected to be the reference satellite. In contrast, if none of the satellites can satisfy this condition, the operation of this round would be terminated.

2) Computation of the Weight Matrix: Because of (9), each ε^i is assumed to be uncorrelated. Accordingly, the weight matrix $\mathbf{W}$ can be simplified as

$$\mathbf{W} = \mathrm{diag}\left(\frac{1}{(\sigma^1)^2}, \ldots, \frac{1}{(\sigma^n)^2}\right) \tag{10}$$

where $\mathrm{diag}(\cdot)$ denotes a diagonal matrix, and σ^i is the standard deviation of ε^i. Note that, although σ^i cannot be quantitatively measured, the CNR values from satellite i to both receivers a and b (ϕ_a^i and ϕ_b^i) can be obtained. Assume that the noise is reversely proportional to the CNR (in decibel-hertz), the variance can be represented by

$$(\sigma^i)^2 \propto 1/\left(\phi_a^i\right)^2 + 1/\left(\phi_b^i\right)^2. \tag{11}$$

Accordingly, the weight matrix $\mathbf{W}$ is obtained by

$$\mathbf{W} = \mathrm{diag}\left(\frac{\left(\phi_a^1\right)^2 \cdot \left(\phi_b^1\right)^2}{\left(\phi_a^1\right)^2 + \left(\phi_b^1\right)^2}, \ldots, \frac{\left(\phi_a^n\right)^2 \cdot \left(\phi_b^n\right)^2}{\left(\phi_a^n\right)^2 + \left(\phi_b^n\right)^2}\right). \tag{12}$$

3) Selection of the Candidate Satellites: Recall that (8) is applicable only when there are at least four satellites shared by the two receivers. Therefore, in addition to the reference satellite, at least another three satellites are expected to be selected, which are referred to as the *candidate satellite*. To prevent the use of pseudorange measurements with very high noise, $\mathrm{CNR}_{\min}$ is defined as another threshold to select candidate satellites. Specifically, only the satellites with higher CNR values than $\mathrm{CNR}_{\min}$ can be selected. In case that the number of candidate satellites is less than three, the operation would be aborted in this round. Note that, since the reference satellite is expected to have the highest CNR value, it requires that $\mathrm{CNR}_{\min} \leq \mathrm{CNR}_{\mathrm{ref}}$.

C. Field Experiments

We conduct a series of field experiments to examine the performance of WLS-DD. The GPS with SiRFstar IV chip is adopted for experiments. The GPS data are collected by using the Telit Evaluation Kit (EVK2) with the GE863-GPS module, and it is logged by using the software SiRFDemo. In the experiment, two GPS receivers are placed apart from each other with certain predetermined distance, called the *baseline*. Each testing lasts for 15–20 min. To compute the distance, we extract the satellite positions, estimated GPS time, and CNRs of all pseudorange measurements from the GPS devices. Specifically, based on the estimated GPS time, we take out the measurements obtained at the same time from the two GPS devices for processing. In addition, the pseudorange measurements are selected according to the predefined thresholds (e.g., $\mathrm{CNR}_{\min} = 30$ dB-Hz and $\mathrm{CNR}_{\mathrm{ref}} = 41$ dB-Hz). Finally, the distance between the two GPS receivers is computed based on (8). For comparison purposes, we also show the distance obtained from the GPS fixes with the same set of selected measurements.

The first set of experiments is conducted in open areas, where GPS receivers have a clear view of the sky. We call this environment as *open_space*. As shown in Fig. 2, the performance is evaluated under different baselines (i.e., 3, 10, 40, and 100 m). In each experiment, around 1200 valid examples are obtained. Across all these examples, we observe that the distances computed by WLS-DD are much closer to the baseline compared with the distances obtained from GPS fixes. This set of experiments demonstrate the stability of WLS-DD under different scales of distance detection. Table I shows the average distance error, which is calculated by $(\sum_{i=1}^{M} |d_i - d_B|/M)$, where M is the total number of valid samples, d_i is the computed distance, and d_B is the baseline distance. As shown, WLS-DD achieves a decent average error of around half a meter in *open_space*.

Since the noise of GPS measurements varies with environments, in addition to the *open_space*, two more typical roadside scenarios are tested. One scenario is the roadside surrounded with trees, which is referred to as the *roadside_trees* environment. Another scenario is the roadside surrounded with buildings, which is referred to as the *roadside_buildings* environment. The baseline is set to 3 m in this set of experiments. As shown in Fig. 3, the performance of both WLS-DD and GPS fixes drops to varying degrees compared with the results obtained in open spaces. This indicates that obstacles such as buildings and trees cause higher noise of GPS measurements. We note that the errors of GPS fixes may go wrong as far as tens of meters. In contrast, WLS-DD achieves much higher accuracy in most cases. In particular, as shown in Table II, the average performance of WLS-DD is around 3–7 times better than that of GPS fixes. It is worth noting that there are many uncertainties that may contribute to the noise in different environments. Therefore, this set of experiments is only used to demonstrate the superiority of WLS-DD, rather than indicating which environment may have a better GPS signal.

IV. Cooperative Vehicular Localization

We propose a cooperative vehicular localization solution with a DSRC-assisted implementation method.

A. Problem Statement

The set of vehicles is denoted $V = \{v_1, v_2, \ldots, v_{|V|}\}$, where $|V|$ is the total number of vehicles. The true location of vehicle v_i $(1 \leq i \leq |V|)$ is represented by the vector $\mathbf{v}_i$. The GPS fix of v_i is denoted $\tilde{\mathbf{v}}_i$. v_j is considered the *neighbor* of v_i if the two vehicles are within their communication range. The set of v_i's neighbors is denoted N_i $(N_i \subset V)$. The true distance between v_i and v_j is denoted d_{ij} (i.e., $d_{ij} = \|\mathbf{v}_i - \mathbf{v}_j\|$), whereas the computed distance (by WLS-DD) between v_i and v_j is denoted $\tilde{d}_{ij}$.

The overall objective of the problem is to estimate vehicle locations by leveraging the GPS fix and the computed interve-

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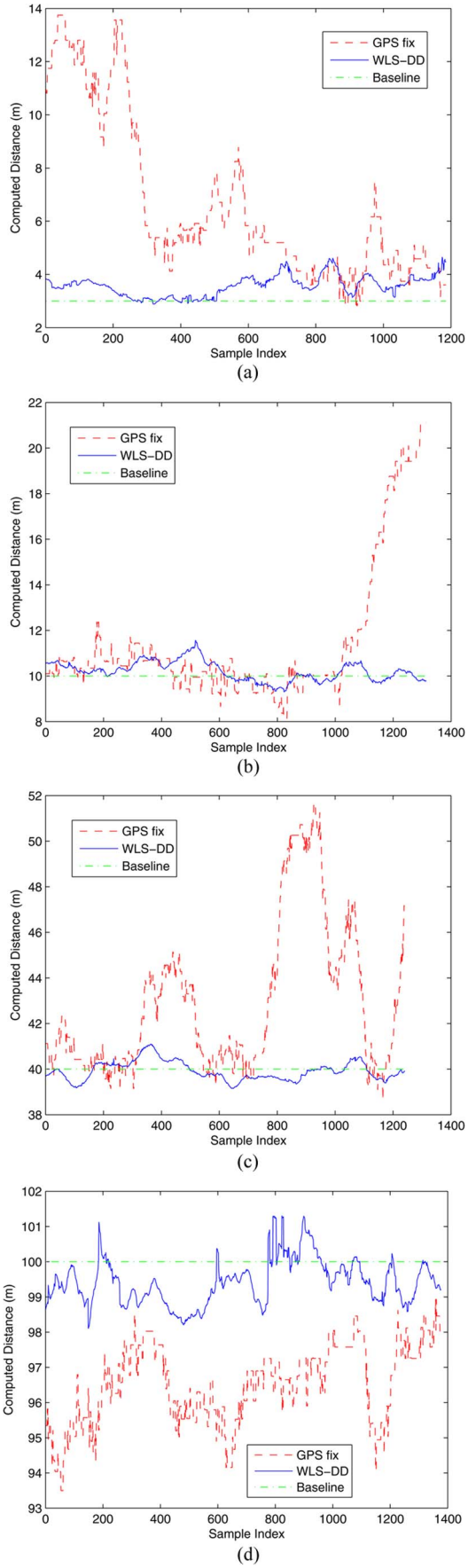


Fig. 2. Distance detection under different baselines. (a) Baseline = 3 m. (b) Baseline = 10 m. (c) Baseline = 40 m. (d) Baseline = 100 m.

TABLE I
AVERAGE ERRORS (M) UNDER DIFFERENT BASELINES

	GPS Fix	WLS-DD
Baseline = 3 m	3.61	0.57
Baseline = 10 m	1.79	0.38
Baseline = 40 m	3.10	0.36
Baseline = 100 m	3.48	0.74

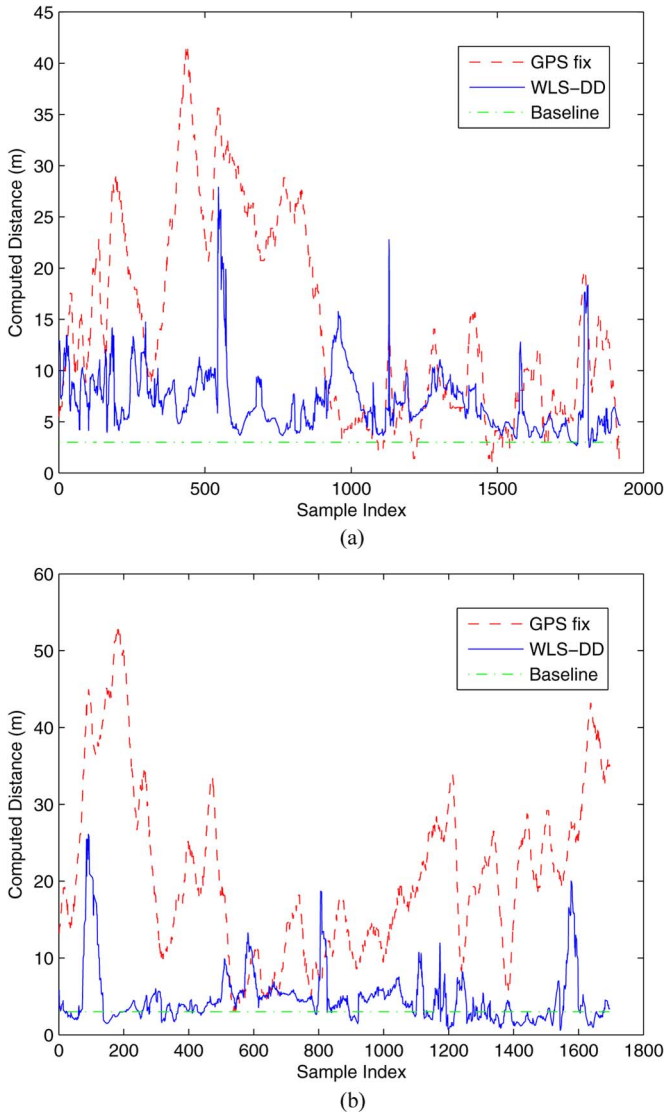


Fig. 3. Distance detection in different environments. (a) Roadside_trees. (b) Roadside_buildings.

TABLE II
AVERAGE ERRORS (M) IN DIFFERENT ENVIRONMENTS

	GPS Fix	WLS-DD
open_space	3.61	0.57
roadside_trees	11.48	4.06
roadside_buildings	17.75	2.31

hicle distance. Specifically, given a set of vehicles V with their GPS fixes $\tilde{\mathbf{v}} = \{\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \ldots, \tilde{\mathbf{v}}_{|V|}\}$, and the computed distance $\tilde{d}_{ij}$ between any two neighboring vehicles v_i and v_j ($v_i \in V$ and $v_j \in N_i$), it is expected to compute a set of estimated locations $\hat{\mathbf{v}} = \{\hat{\mathbf{v}}_1, \hat{\mathbf{v}}_2, \ldots, \hat{\mathbf{v}}_{|V|}\}$, such that each estimated location $\hat{\mathbf{v}}_i$ will be as close as possible to the corresponding true location $\mathbf{v}_i$. In brief, the objective is to minimize the value of

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TABLE III
SUMMARY OF NOTATIONS

Notations	Descriptions	Notes
V	total set of vehicles	$V=\{v_1,v_2,...,v_{\|V\|}\}$
N_i	neighbor set of v_i	$N_i \subset V$
V^k	subset of vehicles with the pivot v_k	$V^k=N_k+\{v_k\}$ and $V^k \subseteq V$
$\mathbf{v}_i$	true location of v_i	
d_{ij}	true distance between v_i and v_j	$d_{ij}=\|\mathbf{v}_i-\mathbf{v}_j\|$
$\tilde{\mathbf{v}}_i$	original GPS fix of v_i	
$\tilde{d}_{ij}$	computed distance between v_i and v_j	computed by WLS-DD
$\boldsymbol{\delta}_i$	GPS error of v_i	$\boldsymbol{\delta}_i=\mathbf{v}_i-\tilde{\mathbf{v}}_i$
$\hat{\mathbf{v}}_i^k$	tentative estimated location of v_i in subset V^k	
$\hat{\mathbf{v}}_i$	final estimated location of v_i	
$\hat{\mathbf{V}}^k$	set of tentative estimated locations computed by v_k	
w_i	weight of the pivot v_i	

$\sum_{i=1}^{|V|}\|\hat{\mathbf{v}}_i-\mathbf{v}_i\|$, which is the overall error of the estimated locations.

B. Proposed Localization Algorithm

We propose the DLEA for cooperative positioning. In general, the DLEA can be divided into two parts, namely, *tentative location estimate* and *final location estimate*. The primary notations used for algorithm design are summarized in Table III.

1) Tentative Location Estimate: To compute tentative estimated locations, each vehicle has to cooperate with its neighbors by sharing their intervehicle distances and the GPS fixes. In this regard, we define the *pivot vehicle* as follows.

Definition 1: The pivot vehicle is defined as follows. Denote V^k as a subset of vehicles, which is constituted by the vehicle v_k and all of its neighbors $\{v_j|v_j \in N_k\}$. Then, v_k is regarded as the pivot vehicle in V^k, where $V^k=N_k+\{v_k\}$.

Given a pivot vehicle v_k in the corresponding set V_k, v_k is supposed to compute a set of tentative estimated locations $\hat{\mathbf{V}}^k=\{\hat{\mathbf{v}}_i^k|v_i \in V^k\}$, where $\hat{\mathbf{v}}_i^k$ represents the tentative estimated location for v_i $(v_i \in V^k)$ computed by v_k. In the following, we present the computation of $\hat{\mathbf{V}}^k$ by formulating a constrained nonlinear optimization problem.

Given v_i $(v_i \in V^k)$ with its GPS fix $\tilde{\mathbf{v}}_i$, the GPS error $\boldsymbol{\delta}_i$ is represented by $\boldsymbol{\delta}_i=\mathbf{v}_i-\tilde{\mathbf{v}}_i$, where $\mathbf{v}_i$ is the true location of v_i. Denote the probability density function (pdf) of the GPS error $\boldsymbol{\delta}_i$ as $\varphi(\boldsymbol{\delta}_i)$; then, $\boldsymbol{\delta}_i$ is most likely to be distributed around the point where the highest probability density is achieved. Therefore, to maximize the possibility that the tentative estimated location $(\hat{\mathbf{v}}_i^k)$ would approximate to the true location $(\mathbf{v}_i)$, it expects to maximize the function of $\varphi(\hat{\mathbf{v}}_i^k-\tilde{\mathbf{v}}_i)$. Moreover, for any $\hat{\mathbf{v}}_i^k$ $(i \neq k)$, it should satisfy the distance constraint between v_i and v_k. That is, given the computed distance $\tilde{d}_{ik}$, it should satisfy that $\|\hat{\mathbf{v}}_i^k-\hat{\mathbf{v}}_k^k\|=\tilde{d}_{ik}$, $\forall\, v_i \in N_k$. Moreover, the road space constraint (e.g., the boundary coordinates), if applicable, can be also included as a constraint for location estimate. To sum up, the set of tentative estimated locations $\hat{\mathbf{V}}^k$ is computed by

$$\hat{\mathbf{V}}^k=\arg\max_{\hat{\mathbf{V}}^k}\sum_{v_i\in V^k}\varphi\left(\hat{\mathbf{v}}_i^k-\tilde{\mathbf{v}}_i\right). \tag{13}$$

Subject to that

$$\left\|\hat{\mathbf{v}}_i^k-\hat{\mathbf{v}}_k^k\right\|=\tilde{d}_{ik} \quad \hat{\mathbf{v}}_i^k \in \mathcal{S}, \qquad \forall\, v_i \in V^k$$

where

- V^k subset with the pivot vehicle v_k;
- $\hat{\mathbf{v}}_i^k$ tentative estimated location for v_i computed by v_k;
- $\tilde{\mathbf{v}}_i$ GPS fix of v_i;
- $\tilde{d}_{ik}$ computed distance between v_i and v_k;
- N_k neighbor set of v_k;
- $\mathcal{S}$ road space constraint.

2) Final Location Estimate: After each vehicle v_k acting as a pivot and computing a set of tentative estimated locations, each v_k obtains $|V^k|$ tentative estimated locations, including the one computed by itself (i.e., $\hat{\mathbf{v}}_k^k$), and another $|V^k|-1$ (i.e., $\{\hat{\mathbf{v}}_k^i|v_i \in N_k\}$) computed by each of its neighbors v_i (when v_i acts as a pivot). Intuitively, given a pivot v_k, the more vehicles that are within V^k, the more likely that $\hat{\mathbf{v}}_i^k$ would be closer to the true location of v_i, because more constraints (i.e., intervehicle distances) are applied in the position estimate. In view of this, we define the *weight of a pivot vehicle* as follows.

Definition 2: The weight of a pivot vehicle is as follows. Given a pivot v_k with its corresponding subset V^k, the weight of v_k, which is denoted w_k, is defined as the number of vehicles in V^k. That is, $w_k=|V^k|$.

The final estimated location $\hat{\mathbf{v}}_k$ is a weighted average of all the tentative estimated locations for v_k $(\{\hat{\mathbf{v}}_k^k\}+\{\hat{\mathbf{v}}_k^i|v_i \in N_k\})$; thus, the set is equivalent to $\{\hat{\mathbf{v}}_k^s|v_s \in V^k\}$, where $\hat{\mathbf{v}}_k^s$ is the tentative estimated location for v_k computed by the pivot v_s. The weight of $\hat{\mathbf{v}}_k^s$ is defined as follows.

Definition 3: The weight of a tentative estimated location is as follows. Given the tentative estimated location for v_k computed by the pivot v_s $(\hat{\mathbf{v}}_k^s)$, the weight of $\hat{\mathbf{v}}_k^s$ is defined as the weight of v_s over the sum of the weight of each v_i in the subset V^k, which is computed by $(w_s/\sum_{v_i\in V^k}w_i)$.

Finally, $\hat{\mathbf{v}}_k$ is a linear combination of all the tentative estimated locations $\hat{\mathbf{v}}_k^s$, which is computed by

$$\hat{\mathbf{v}}_k=\sum_{v_s\in V^k}\frac{w_s}{\sum\limits_{v_i\in V^k}w_i}\cdot\hat{\mathbf{v}}_k^s. \tag{14}$$

The main procedures of DLEA are summarized as follows.

- Compute the distance between any two neighboring vehicles based on WLS-DD.
- Each vehicle v_i shares its GPS fix $\tilde{\mathbf{v}}_i$ with its neighbors.
- Each vehicle v_i acts as a pivot and computes a set of tentative estimated locations $\hat{\mathbf{V}}^i$.
- Each vehicle v_i shares the computed $\hat{\mathbf{V}}^i$ with its neighbors.
- Each vehicle v_i computes its final estimated location $\hat{\mathbf{v}}_i$.

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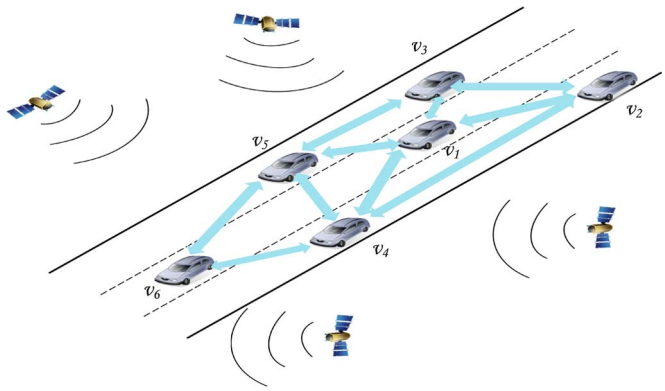


Fig. 4. Example of cooperative localization.

C. Example

Fig. 4 gives an example to better illustrate the cooperative localization via DLEA. There is a set of vehicles $V = \{v_1, v_2, v_3, v_4, v_5, v_6\}$ in the given road space $\mathcal{S}$. The edge with double arrows represents that the two vehicles are within their communication range. Take v_1 as an example, the neighbor set of v_1 is represented by $N_1 = \{v_2, v_3, v_4, v_5\}$, and the subset of vehicles with pivot v_1 is $V^1 = \{v_1, v_2, v_3, v_4, v_5\}$. The set of tentative estimated locations $\hat{\mathbf{V}}^1 = \{\hat{\mathbf{v}}_1^1, \hat{\mathbf{v}}_2^1, \hat{\mathbf{v}}_3^1, \hat{\mathbf{v}}_4^1, \hat{\mathbf{v}}_5^1\}$ is computed by

$$\hat{\mathbf{V}}^1 = \arg\max_{\hat{\mathbf{V}}^1} \left(\varphi\left(\hat{\mathbf{v}}_1^1 - \tilde{\mathbf{v}}_1\right) + \varphi\left(\hat{\mathbf{v}}_2^1 - \tilde{\mathbf{v}}_2\right) \cdots + \varphi\left(\hat{\mathbf{v}}_5^1 - \tilde{\mathbf{v}}_5\right)\right). \tag{15}$$

Subject to that

$$\left\|\hat{\mathbf{v}}_2^1 - \hat{\mathbf{v}}_1^1\right\| = \tilde{d}_{21}, \left\|\hat{\mathbf{v}}_3^1 - \hat{\mathbf{v}}_1^1\right\| = \tilde{d}_{31}, \left\|\hat{\mathbf{v}}_4^1 - \hat{\mathbf{v}}_1^1\right\| = \tilde{d}_{41},$$
$$\left\|\hat{\mathbf{v}}_5^1 - \hat{\mathbf{v}}_1^1\right\| = \tilde{d}_{51}, \quad \text{and} \quad \hat{\mathbf{v}}_i^1 \in \mathcal{S}, \qquad \forall\, v_i \in V^1.$$

For clear exposition, we consider in the 2-D coordinate. Nevertheless, it is straightforward to extend the computation into the 3-D coordinate. Assume that the error $\boldsymbol{\delta}$ follows the Gaussian distribution ($\boldsymbol{\delta} \sim N(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$). Then, the pdf is

$$\varphi(\boldsymbol{\delta}) = \frac{1}{\sqrt{2\pi}\boldsymbol{\sigma}} \cdot e^{-\frac{(\boldsymbol{\delta}-\boldsymbol{\mu})^2}{2\boldsymbol{\sigma}^2}} \tag{16}$$

where $\boldsymbol{\mu}$ is the mean, and $\boldsymbol{\sigma}^2$ is the variance. For the vehicle v_i in an xy coordinate, its tentative estimated location $\hat{\mathbf{v}}_i^k$, GPS fix $\tilde{\mathbf{v}}_i$, and localization error $\boldsymbol{\delta}_i$ can be expanded to $\hat{\mathbf{v}}_i^k(\hat{v}_{i_x}^k, \hat{v}_{i_y}^k)$, $\tilde{\mathbf{v}}_i(\tilde{v}_{i_x}, \tilde{v}_{i_y})$, and $\boldsymbol{\delta}_i(\delta_{i_x}, \delta_{i_y})$, respectively. Accordingly, the objective function is decomposed into $\varphi(\hat{v}_{i_x}^k - \tilde{v}_{i_x}) \cdot \varphi(\hat{v}_{i_y}^k - \tilde{v}_{i_y})$, and the set of tentative estimated locations is computed by

$$\hat{\mathbf{V}}^k = \arg\max_{\hat{\mathbf{V}}^k} \sum_{v_i \in V^k} \varphi\left(\hat{v}_{i_x}^k - \tilde{v}_{i_x}\right) \cdot \varphi\left(\hat{v}_{i_y}^k - \tilde{v}_{i_y}\right). \tag{17}$$

Based on (16), the final formation of $\hat{\mathbf{V}}^k$ is

$$\hat{\mathbf{V}}^k = \arg\min_{\hat{\mathbf{V}}^k} \sum_{v_i \in V^k} \frac{\left(\hat{v}_{i_x}^k - \tilde{v}_{i_x} - \mu_x\right)^2}{2\sigma_x^2} + \frac{\left(\hat{v}_{i_y}^k - \tilde{v}_{i_y} - \mu_y\right)^2}{2\sigma_y^2} \tag{18}$$

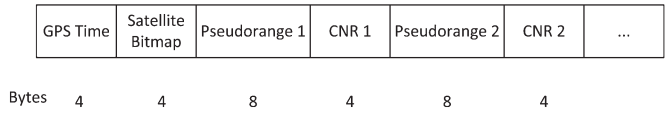


Fig. 5. Message format for disseminating pseudorange information.

where μ_x and μ_y are the mean values, whereas σ_x and σ_y are the standard deviations in the xy coordinate. Subject to that are the following:

$$\sqrt{\left(\hat{v}_{j_x}^k - \hat{v}_{k_x}^k\right)^2 + \left(\hat{v}_{j_y}^k - \hat{v}_{k_y}^k\right)^2} = \tilde{d}_{jk} \qquad \forall\, v_j \in N_k, \text{ and}$$
$$S_{\text{lb}}^x \le \hat{v}_{i_x}^k \le S_{\text{ub}}^x, \quad S_{\text{lb}}^y \le \hat{v}_{i_y}^k \le S_{\text{ub}}^y \qquad \forall\, v_i \in V^k.$$

Detailed derivation of (18) can be found in [21]. When the geographical bound is available, S_{lb}^x, S_{ub}^x, S_{lb}^y, and S_{ub}^y are used to define the road space constraint. That is, they represent the lower and upper bounds for xy coordinates of the road space, respectively. Note that these coordinate constraints are optional parameters for the DLEA.

With the computation of all the sets of tentative estimated locations, the DLEA computes the final estimate location for each vehicle. For example, considering v_6 as a pivot vehicle shown in Fig. 4, its subset is $V^6 = \{v_4, v_5, v_6\}$. Therefore, v_6 receives three tentative estimated locations in the first step, namely, $\hat{\mathbf{v}}_6^4$, $\hat{\mathbf{v}}_6^5$, and $\hat{\mathbf{v}}_6^6$. According to Definition 2, the weights of v_4, v_5, and v_6 are $w_4 = 4$, $w_5 = 4$, and $w_6 = 2$, respectively. Therefore, we have $\sum_{v_i \in V^6} w_i = 10$. Based on Definition 3, the weights of $\hat{\mathbf{v}}_6^4$, $\hat{\mathbf{v}}_6^5$, and $\hat{\mathbf{v}}_6^6$ will be 4/10, 4/10, and 2/10, respectively. Therefore, the final estimated location for v_6 is computed by $\hat{\mathbf{v}}_6 = ((4/10) \cdot \hat{\mathbf{v}}_6^4 + (4/10) \cdot \hat{\mathbf{v}}_6^5 + (2/10) \cdot \hat{\mathbf{v}}_6^6)$.

D. DSRC-Assisted Implementation

Timely and efficient data dissemination is critical to enabling innovative applications in vehicular networks [22]–[24], whereas DSRC has shown great potential in supporting cooperative positioning [8]. In the following, we analyze the feasibility of applying DSRC to the proposed solution. The basic safety message (BSM), as defined in SAE J2735 [25], is suitable for data sharing among neighboring vehicles. The broadcast period of the BSM is specified to 100 ms. In the process of computing intervehicle distances with WLS-DD, the periodical pseudorange measurements are piggybacked into the BSM. The format of the message is shown in Fig. 5, The 4-B satellite bitmap field is made up of 32 ON/OFF flags, indicating the presence of the pseudorange measurement of the corresponding satellite in the latter section of the message. Each measured pseudorange is represented by an 8-B double-precision floating-point data type. When there are n candidate satellites, the total length of the message is $8 + 12 \times n$ B. For example, with the minimum requirement (four satellites), the attached message is 56 B. The control channel as defined in IEEE 1609.4 [26] is adopted for broadcasting BSMs.

After obtaining the intervehicle distances, each vehicle computes a set of tentative estimated locations. This information, together with the GPS fix, are supposed to be shared among neighboring vehicles. According to the specification in SAE J2735, the GPS fix has already been included in the 39-B

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Num of Neighbors	Neighbor ID	Latitude	Longitude	Elevation	Neighbor ID	...
Bytes 4	4	4	4	2	4	

Fig. 6. Message format for disseminating tentative estimated locations.

payload of the BSM. Therefore, only the tentative estimated locations need to be padded. The designed message format is shown in Fig. 6. The data type for representing vehicle locations is specified in SAE J2735. That is, a 4-B data type are used for latitude and longitude, and a 2-B data type is used for elevation. Accordingly, given m neighboring vehicles, the message length is $4 + 14 \times m$ B. Considering a heavy traffic scenario with high vehicle density (e.g., $m = 100$), the message length is 1404 B. Together with the 39-B payload of the BSM, the total packet length is 1443 B. According to DSRC, the data rate of 6 Mb/s is recommended for disseminating safety critical messages with high reliability. Therefore, the transmission time is less than 2 ms (1443 $\times$ 8 bits/6 Mb/s). In addition, we consider that only the vehicles moving in the same direction will participate into the cooperative positioning. Therefore, given the low relative speeds among the vehicles and the short transmission latency, it is feasible to adopt DSRC to enable the proposed solution.

V. Data Fitting

To examine the effect of intervehicle distance detection (e.g., using WLS-DD) to the positioning performance (e.g., using DLEA), we derive a data fitting model to determine a probability distribution with appropriate parameters based on the real observed data from the field experiments. Since GPS errors are caused by many independent sources, such as satellite clock bias, atmospheric delay, acquisition noise, and multipath, etc., it is commonly assumed that the error follows the Gaussian distribution [5], [13], [27]. Accordingly, we estimate the parameters by fitting the observed data into the Gaussian distribution. Specifically, the maximum likelihood estimation [28] is adopted, and the detailed procedures are presented as follows.

Given the set of observed distance errors $\delta_d = (\delta_{d_1}, \delta_{d_2}, \ldots, \delta_{d_n})$, denote $f(\delta_d|\mu_d, \sigma_d^2)$ as the pdf, where μ_d is the mean, and σ_d^2 is the variance. Suppose that the set of measurements is independent and identically distributed; then, the joint pdf can be expressed as follows:

$$f\left(\delta_d = (\delta_{d_1}, \delta_{d_2}, \ldots, \delta_{d_n})\,|\mu_d, \sigma_d^2\right) = \prod_{i=1}^{n} f\left(\delta_{d_i}|\mu_d, \sigma_d^2\right). \tag{19}$$

With the Gaussian distribution, the likelihood function $L(\mu_d, \sigma_d^2|\delta_d)$ is computed by

$$\begin{aligned} L\left(\mu_d, \sigma_d^2|\delta_d\right) &= \prod_{i=1}^{n} f\left(\delta_{d_i}|\mu_d, \sigma_d^2\right) \\ &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma_d} e^{-\frac{\left(\delta_{d_i}-\mu_d\right)^2}{2\sigma_d^2}} \\ &= \left(2\pi\sigma_d^2\right)^{-\frac{n}{2}} \cdot e^{-\frac{\sum_{i=1}^{n}\left(\delta_{d_i}-\mu_d\right)^2}{2\sigma_d^2}}. \end{aligned} \tag{20}$$

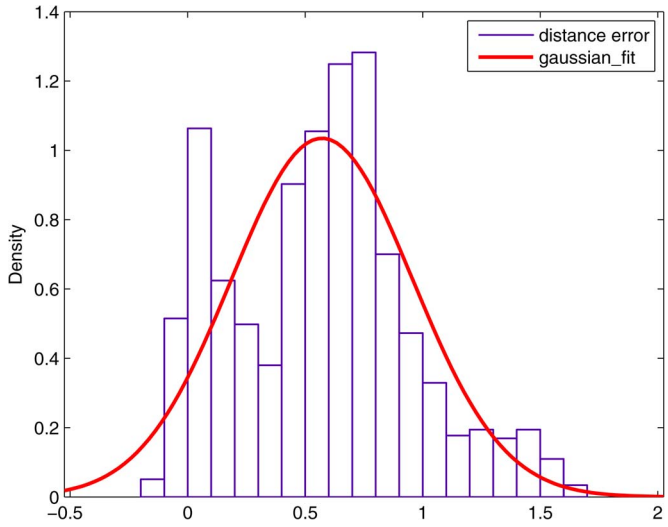


Fig. 7. Example of data fitting.

The estimated parameters (i.e., the mean value and the variance), which are denoted $\hat{\mu}_d$ and $\hat{\sigma}_d^2$, are supposed to maximize the value of the likelihood function so that they give the observed data the greatest probability. For computational convenience, the log-likelihood function is adopted by taking the logarithm of $L(\mu_d, \sigma_d^2|\delta_{d_1}, \delta_{d_2}, \ldots, \delta_{d_n})$, and the estimated parameters are obtained by solving the following:

$$\begin{cases} \frac{\partial \ln L\left(\mu_d, \sigma_d^2|\delta_{d_1}, \delta_{d_2}, \ldots, \delta_{d_n}\right)}{\partial \mu_d} = \frac{1}{\sigma_d^2}\sum_{i=1}^{n}\left(\delta_{d_i} - \mu_d\right) = 0 \\ \frac{\partial \ln L\left(\mu_d, \sigma_d^2|\delta_{d_1}, \delta_{d_2}, \ldots, \delta_{d_n}\right)}{\partial \sigma_d^2} = -\frac{n}{2\sigma_d^2} + \frac{1}{2\sigma_d^4}\sum_{i=1}^{n}\left(\delta_{d_i} - \mu_d\right)^2 = 0. \end{cases}$$

The solutions are represented by

$$\begin{cases} \hat{\mu}_d = \sum_{i=1}^{n} \delta_{d_i}/n \\ \hat{\sigma}_d^2 = \frac{1}{n}\sum_{i=1}^{n}\left(\delta_{d_i} - \hat{\mu}_d\right)^2 \end{cases} \tag{21}$$

where δ_{d_i} $(1 \le i \le n)$ is the ith sample of distance error, and n is the total number of samples.

We give an example based on the field experimental results shown in Fig. 2(a) to show that this fitting model is suitable for deriving parameters to evaluate the algorithm performance. There are 1185 distance samples obtained by WLS-DD ($n = 1185$). Denote each sample as s_{d_i} $(1 \le i \le n)$. Then, the distance error δ_{d_i} is computed by $\delta_{d_i} = |s_{d_i} - baseline|$, where baseline = 3 m in this example. With the obtained data set $\delta_d = (\delta_{d_1}, \delta_{d_2}, \ldots, \delta_{d_n})$, the estimated parameters are computed by (21), which are $\hat{\mu}_d = 0.57$ and $\hat{\sigma}_d^2 = 0.14$. We check the performance of this fitting by plotting a histogram of the observed data set and superimposing the pdf with the estimated parameters. The histogram is constructed according to the Scott's rule [29]. The fitting result is shown in Fig. 7. Comparing the curve of Gaussian fitting and the data histogram, by and large, the derived parameters are able to capture the true distribution of errors. It is worth noting that the Gaussian fitting may not be the best model to fit the observed errors. More sophisticated models may give better fitting results. Although different fitting models

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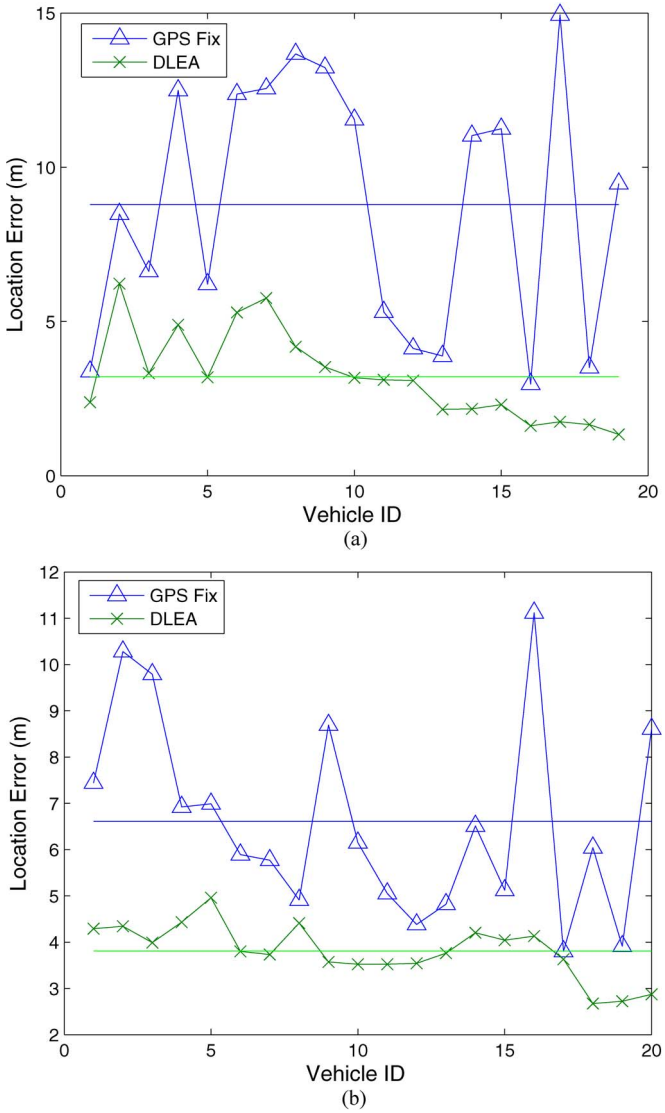


Fig. 8. Performance comparison by using different sources of inputs. (a) Using distance errors from filed experiments as inputs. (b) Using distance errors from the Gaussian fitting as inputs.

may derive different parameters used as the simulation input for performance evaluation, we claim that the relative performance of the compared algorithms (i.e., DLEA and GPS fixes) shall remain the same, as long as the fitting model is able to properly capture the statistical characteristics of the observed errors.

To verify this point, we set up a control experiment as follows. The two experiments are conducted under the same default setting (to be elaborated in Section VI), except that the first experiment uses the observed distance errors from the *open_space* environment as the simulation input. The corresponding result is shown in Fig. 8(a). In contrast, the second experiment uses the estimated parameters for *open_space* to generate distance errors, and these errors are used as the simulation input. The corresponding result is shown in Fig. 8(b). We note that the relative performance of GPS fixes and DLEA is consistent in these two settings. Moreover, the mean localization error of each algorithm is represented by the straight line in the figure. As shown, the mean localization errors of GPS fixes in these two settings are 8.79 and 6.61 m, respectively. On the other hand, the mean localization errors of DLEA in these two settings are 3.21 and 3.81 m, respectively. Clearly, both algorithms have similar average performance when using different sets of error inputs. The given analysis demonstrates conclusively that it is reasonable to adopt this Gaussian fitting model to derive parameters for comparing the performance between GPS fixes and DLEA.

VI. Performance Evaluation

The simulation model is built by CSIM19 [30]. The DLEA is implemented by C programming together with MATLAB. In particular, the main framework of DLEA is implemented by C programming, whereas the constrained nonlinear optimization problem is approximately solved by the active-set method [31] using MATLAB. We simulate a road snapshot where the lane length is 200 m and the lane width is 3 m, which are adopted as the coordinate bounds as formulated in the optimization problem. The true locations of vehicles spatially form a Poisson process with the mean intervehicle distance of 10 m. The error of GPS fixes follows the Gaussian distribution with zero mean and deviation of 15 m. The error of intervehicle distance measurements follows the Gaussian distribution with zero mean and deviation of 5 m. The communication range among vehicles is set to 100 m, which has been verified as a conservative data transmission distance supported by DSRC [32]. Unless stated otherwise, the simulations are conducted under these default settings. Each simulation is repeated for ten times with different seeds, and the simulation results are obtained by taking the mean values.

A. Effect of Distributed Localization

DLEA computes the location of each vehicle only requiring its one-hop neighbors' information. For comparison purposes, we consider a *centralized solution*, which computes the location of each vehicle by collecting the information (i.e., the GPS fixes and the intervehicle distances) of all the other vehicles. The operation of the centralized solution is similar with DLEA, except the assumption that the global information is available for each vehicle. Note that there is no predefined road length in this setting due to the various numbers of simulated vehicles. Fig. 9 compares the average localization errors of the centralized solution, DLEA, and GPS fixes. As shown, DLEA achieves the same performance with the centralized solution when there are only a few vehicles (i.e., ten vehicles). This makes sense because, in the default setting, all these ten vehicles are within the communication range of each other, which makes DLEA essentially a centralized solution. With an increasing number of vehicles, as expected, the centralized solution outperforms DLEA. However, compared with GPS fixes, the performance of DLEA is much closer to the centralized solution.

B. Effect of Environments

In Section III-C, we have collected the errors for both GPS fixes and WLS-DD in three different environments, namely, *open_space*, *roadside_trees,* and *roadside_buildings*. In

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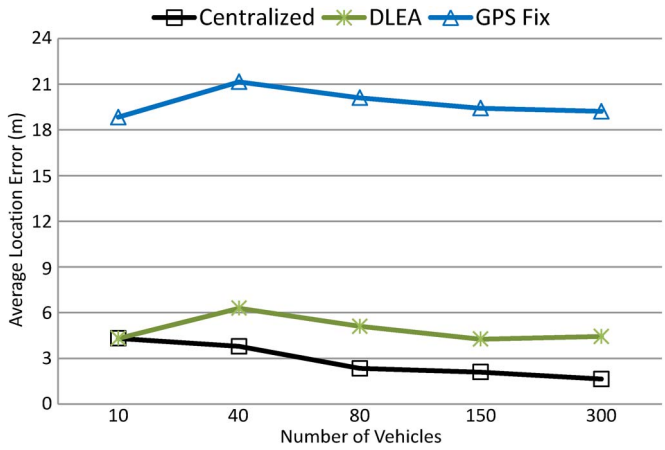


Fig. 9. Performance evaluation by comparing with the centralized solution.

TABLE IV
ESTIMATED PARAMETERS (M) IN DIFFERENT ENVIRONMENTS

	GPS Fix		WLS-DD	
	$\hat{\mu}_{gps}$	$\hat{\sigma}^2_{gps}$	$\hat{\mu}_d$	$\hat{\sigma}^2_d$
open_space	3.61	9.55	0.57	0.15
roadside_trees	11.41	91.58	4.05	10.11
roadside_buildings	17.75	120.56	1.73	13.03

addition, we have derived the parameters (mean and variance in Gaussian distribution) to model these collected errors for both GPS fixes and DLEA, which are summarized in Table IV. In the following simulation, the GPS errors are generated by the estimated parameters for GPS fixes, whereas the intervehicle distance errors are generated by the estimated parameters for WLS-DD. Fig. 10 compares the algorithm performance in different environments [the result for *open_space* has been shown in Fig. 8(b)]. Specifically, the x-axis represents the vehicle IDs, and the y-axis represents the localization errors. Note that the vehicle IDs are generated sequentially in the simulation, and they do not imply any geometric relationship among vehicles. The location errors of GPS and DLEA are computed by $\|\tilde{\mathbf{v}}_i - \mathbf{v}_i\|$ and $\|\hat{\mathbf{v}}_i - \mathbf{v}_i\|$, respectively. As shown, DLEA outperforms GPS fixes in all environments. Table V summarizes different statistics of location errors. The mean errors of GPS fixes and DLEA are computed by $\sum_{v_i \in V} \|\tilde{\mathbf{v}}_i - \mathbf{v}_i\|/|V|$ and $\sum_{v_i \in V} \|\hat{\mathbf{v}}_i - \mathbf{v}_i\|/|V|$, respectively. The 90th percentile error is computed based on linear interpolation between the closest ranks. Moreover, the maximal errors of both GPS fixes and DLEA are collected for comparison. As shown, DLEA significantly outperforms GPS fixes in a variety of circumstances.

C. Effect of Communication Range

Intuitively, when more vehicles can share information with each other and perform cooperative positioning, the estimated location may have higher accuracy. The communication range is one of the factors that may influence the number of vehicles that are able to participate into cooperative positioning. In the following simulation, we set the *roadside_buildings* as the default environment. Fig. 11 examines the effect of communication range on positioning performance. As expected,

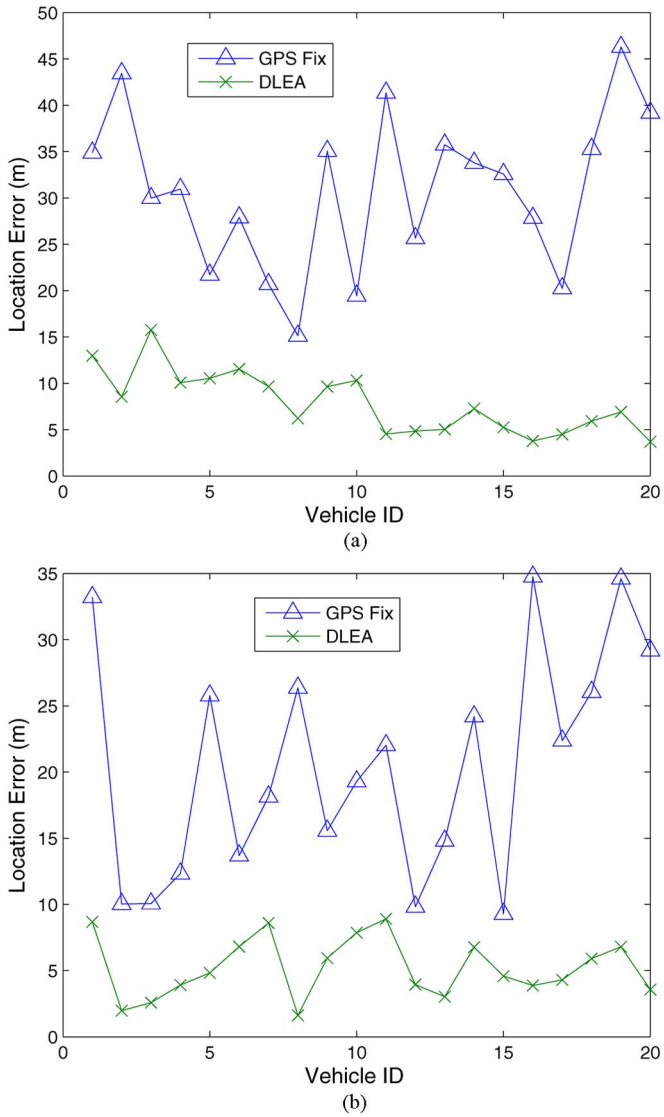


Fig. 10. Performance evaluation in different environments. (a) Roadside_trees. (b) Roadside_buildings.

TABLE V
STATISTICS OF LOCATION ERRORS (M) IN DIFFERENT ENVIRONMENTS

Environment	GPS Fix			DLEA		
	mean	90-percentile	maximal	mean	90-percentile	maximal
open_space	6.61	10.04	11.12	3.81	4.42	4.96
roadside_trees	20.58	33.91	34.75	5.22	8.64	8.91
roadside_buildings	30.85	42.4	46.28	7.84	12.25	15.75

DLEA achieves better performance with an increase in the communication range. Table VI summarizes the statistics of location errors under different communication ranges. Note that GPS fixes do not rely on vehicular communications. Therefore, the performance of GPS fixes stays the same.

D. Effect of Traffic Density

In addition to the communication range, the traffic density is another factor that will affect the number of vehicles participating in cooperative positioning. The following simulation evaluates the performance of DLEA under different traffic densities. Since the road length is 200 m by default, the number

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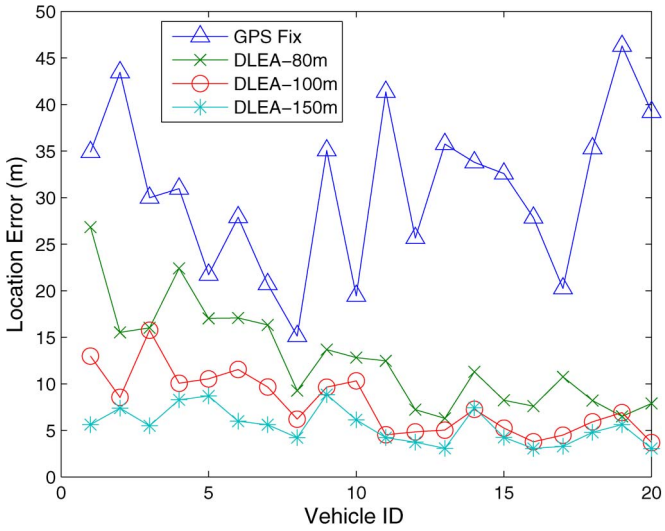


Fig. 11. Performance evaluation under different communication ranges.

TABLE VI
STATISTICS OF LOCATION ERRORS (M) UNDER DIFFERENT COMMUNICATION RANGES

Comm Range	GPS Fix			DLEA		
	mean	90-percentile	maximal	mean	90-percentile	maximal
80 m	30.85	42.4	46.28	12.66	19.74	26.83
100 m	30.85	42.4	46.28	7.84	12.25	15.75
150 m	30.85	42.4	46.28	5.43	8.49	8.83

of simulated vehicles shall vary with the change of traffic densities. For example, there are 40 vehicles in Fig. 12(a), but only 13 vehicles in Fig. 12(b) (the vehicle ID always counts from one, and it is a consecutive integer; therefore, the last vehicle ID represents the total number of vehicles). This is because the mean intervehicle distance changes from 5 to 15 m. Table VII summarizes different statistics of location errors. These results demonstrate that DLEA is effective under different traffic densities.

E. Algorithm Implementation Comparison

We compare DLEA with other well-known localization techniques with respect to the implementation cost, including RF-GPS [9], Kalman-filter-based solution [7], and Vision/DGPS-aided INS [18]. The implementation of these cooperative positioning solutions, in one way or another, rely on the support of particular hardware devices or infrastructures. The specific requirements are summarized in Table VIII, where "N" represents that the corresponding device/technique is not required, whereas "Y" represents that it is required. As noted, with respect to the communication medium, all the solutions are implemented based on DSRC, whereas for other requirements, such as reference points, auxiliary devices, and ranging equipment, etc., only DLEA can be implemented without these extra costs. This implies that DLEA is potentially a practical solution. Nevertheless, it is worth noting that there is a tradeoff between the robustness and the implementation cost of solutions. Although DLEA has been demonstrated effective in some environments, it is still not robust enough for safety-critical applications in current status. In this regard, on top of DLEA, the given techniques can be incorporated to further enhance the positioning accuracy and robustness based on the requirement of specific applications.

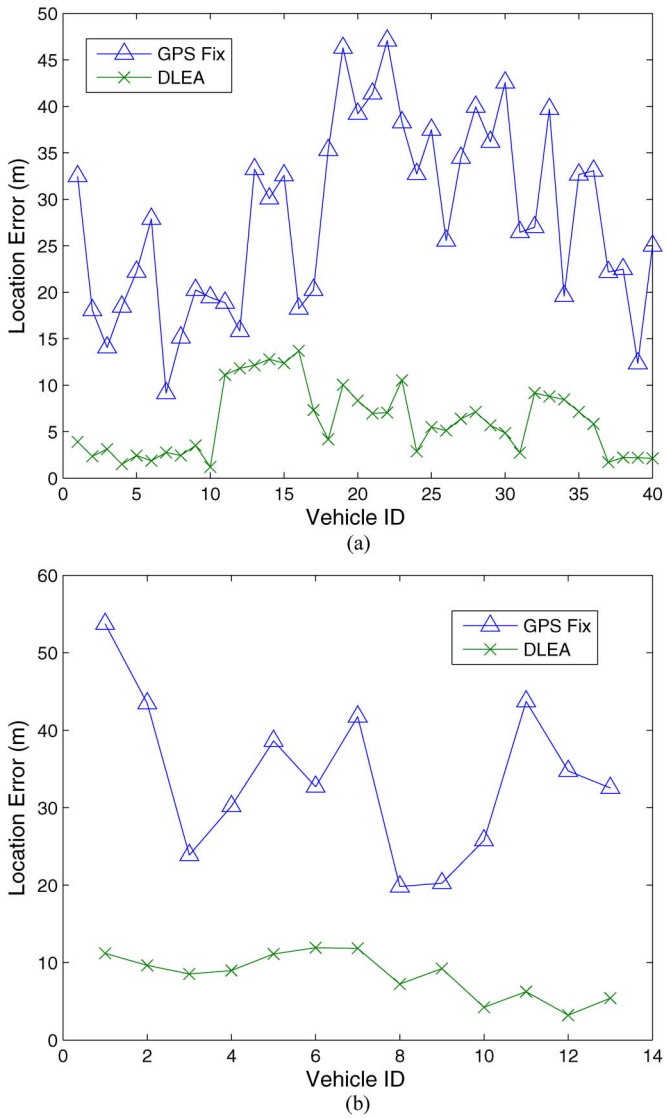


Fig. 12. Performance evaluation under different traffic densities. (a) Average intervehicle distance = 5 m. (b) Average intervehicle distance = 15 m.

TABLE VII
STATISTICS OF LOCATION ERRORS (M) UNDER DIFFERENT TRAFFIC DENSITIES

Inter-vehicle dist	GPS Fix			DLEA		
	mean	90-percentile	maximal	mean	90-percentile	maximal
5 m	28.07	40.66	47.07	6.03	11.97	13.69
10 m	30.85	42.4	46.28	7.84	12.25	15.75
15 m	33.93	45.71	53.71	8.36	11.85	11.91

TABLE VIII
ALGORITHM COMPARISON IN IMPLEMENTATION COST

Algorithm / Supporting Device	DLEA	RF-GPS	Kalman filter based solution	Vision/DGPS-aided INS
Communication medium	DSRC	DSRC and RFID radio	DSRC	DSRC
Reference points (anchors or roadside units with known exact coordinates)	N	Y	N	Y
Auxiliary augmentation devices (inertial sensors, maps, camera, etc.)	N	N	Y	Y
Radio-based ranging with particular equipment (RSSI, ToA, AoA, etc.)	N	N	Y	Y

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VII. Conclusion and Future Work

In this paper, we have focused on improving positioning accuracy in vehicular networks using GPS pseudorange measurements. Two algorithms, namely WLS-DD and DLEA, are proposed to enable the cooperative positioning. Specifically, WLS-DD detects intervehicle distances based on the sharing of GPS pseudorange measurements. Furthermore, it utilizes CNRs to differentiate the accuracy level of different pseudorange measurements so that it can mitigate the impact of random noise in coarse code acquisition. The field experimental results demonstrate that WLS-DD significantly improves the accuracy of distance detection in different environments. Based on WLS-DD, DLEA is proposed to enable cooperative vehicular localization by sharing the GPS fixes and the detected intervehicle distances. With the assistance of DSRC, the implementation of DLEA does not rely on any specialized sensors or infrastructures, which makes it an inexpensive and practical solution. To examine the effect of WLS-DD on the performance of DLEA, the Gaussian fitting model is used to derive parameters based on collected data from the field experiments. We have shown that it is suitable for adopting the derived parameters for performance evaluation. An extensive simulation study demonstrates that the proposed solutions can effectively improve the positioning accuracy under a variety of conditions.

As future directions of this paper, the communication overhead and its impacts on cooperative localization will be further examined. Moreover, more sophisticated solutions (e.g., incorporating with vehicle kinematics and specialized sensors) are expected to be designed and evaluated to enhance positioning performance in practical conditions.

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